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MILITARY SPECIFICATION

BUTTON, INSIGNIA, METAL, UNIFORM AND CAP

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers metal uniform and cap buttons with embossed designs used by the Military Departments.

1.2 Classification. Buttons will be of the following types, styles, classes, subclasses and sized as specified (see 1.2.1, 6.2, 6.5 and 6.6)

Type I, Button, Insignia, Metal Uniform
- Style 1 – Air Force, Silver Plated
  - Class A – Air Force (with coat of arms design)
  - Class B – Air Force (with wings and star design)
- Style 2 – Army
  - Class A – Army, Gold Plated, Solid Design
  - Class B – Army, Gold Plated, Pierced Design
  - Class C – Army, Gold Plated, Corps of Engineers, Pierced Design
  - Class D – Army, Gold Plated, JROTC Division, Solid Design
  - Class E – Army, Nickel Plated, Pierced Design (See 1.2.1 and 6.6)
  - Class F – Army, Brass Plated, Antique Finish, Solid Design (AGSU)
- Style 3 – Marine Corps
  - Class A – Gold Plated
  - Class B – Black Enamel
    - Subclass 1 – High Contour
- Style 4 – Navy
  - Class A – Gold Plated
  - Class B – Bronze
  - Class C – Silver Plated
  - Class D - Nickel Plated, Gold Plated, Mirror-like Polished

Comments, suggestions, or questions on this document should be addressed to The Institute of Heraldry, ATTN: Technical and Production Division, 9325 Gunston Road, Room S-108, Fort Belvoir, VA 22060-5579 or email to usarmy.belvoir.hqda.mil.tioh-webmaster@mail.mil. You may verify the currency of this specification using the ASSIST Online database at https://assist.dla.mil.

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Style 5 – Coast Guard (See 6.3)
Class A – Gold Plated
Class B – Bronze
Style 6 – Space Force
Class A – Silver Plated
Class B – Nickel Plated, Mirror Polished
Type II – Button, Insignia, Metal, Cap
Style 1 – Air Force, Silver Plated
Style 2 – Army
Class A – Army, Gold Plated, Solid Design
Class B – Army, Gold Plated, Pierced Design
Class C – Army, Corps of Engineers, Gold Plated, Pierced Design
Class D – Army, JROTC Division, Gold Plated, Solid Design
Class E – Army, Nickel Plated, Pierced Design (See 1.2.1 and 6.6)
Class F – Army, Brass Plated, Antique Finish, Solid Design (AGSU)
Style 3 – Marine Corps
Class A – Gold Plated
Subclass 1 – High Contour
Class B – Black Enameled
Subclass 1 – High Contour (Officers)
Subclass 2 – Low Contour (Enlisted Personnel)
Style 4 – Navy
Class A – Gold Plated
Class B – Bronze
Class C – Silver Plated
Class D – Nickel Plated, Gold Plated, Mirror Like Polished
Style 5 – Coast Guard (See 6.3)
Class A – Gold Plated
Subclass 1 – Male
Subclass 2 – Female
Class B – Bronze
Style 6 – Coast Guard Auxiliary (See 6.3)
Style 7 – Space Force
Class A – Silver Plated
Class B – Nickel Plated, Mirror Polished
Type III – Button, Insignia, Metal, Cuff Link, Air Force

1.2.1 Sizes. Button sizes will be as shown in table I. Sizes will be given in lignes where 40 lignes equal one inch **1/**.

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TABLE I. Sizes (in Lignes)

ARMY	AIR FORCE	MARINE CORPS	NAVY	COAST GUARD	COAST GUARD AUX	ARMY JROTC	ARMY **2/** SPECIAL	ARMY CORPS OF ENGRS	SPACE FORCE
20	20	--	--	--	--	--	--	--	20
--	--	22	--	--	--	--	--	--	--
--	--	--	22 1/2	22 1/2	22 1/2	--	--	--	--
25	25	--	--	--	--	25	25	25	25
--	--	27	--	--	--	--	--	--	--
--	--	--	28	28	28	--	--	--	--
30	30	--	--	--	--	30	--	30	30
--	--	--	35	35	35	--	--	--	--
36	36	--	--	--	--	36	36	36	36
--	--	40 **1/**	40 **1/**	40 **1/**	--	--	--	--	--
--	45 **3/**	--	--	--	--	--	--	--	--

1/ A tolerance of ± 1 ligne will be permitted.
2/ Army Special buttons are restricted for use by the United States Army Band and Ceremonial Units of the U.S. Army 3d Infantry.
3/ Air Force uses 45 ligne buttons only with Type I, Style 1, Class A buttons.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this document to extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of specifications and standards MIL-STD-961 and supplement thereto, cited in the solicitation (see 6.2).

FEDERAL SPECIFICATIONS

A-A-52094 - Thread, Cotton

MILITARY STANDARDS

MIL-STD-961 – Defense and Program – Unique Specifications format and Content

(Copies of these documents are available online at https://quicksearch.dla.mil).

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

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THE INSTITUTE OF HERALDRY DRAWINGS

4-4-92 - Buttons, Insignia, Metal, Uniform and Cap (also illustrated as Figure 1)
B-6-5 - Button, ROTC, JR Division (also illustrated in Figure 3)
B-6-10 - Buttons, USAF, US Navy, US Coast Guard, and US Marine Corps (also illustrated in Figure 2)
B-6-10A - Button, US Air Force (also illustrated in Figure 4)
B-6-045 – Button, US Space Force (also Illustrated in Figure 5)

(Figures 1, 2, 3, 4, 5 and 6 are miniature reproductions of the referenced drawings and are attached for information only.)

THE INSTITUTE OF HERALDRY U.S. ARMY STANDARD METAL FINISH CHIP SET

(Copies of standards, drawings and Chip sets are available from The Institute of Heraldry, ATTN: Technical and Production Division, 9325 Gunston Road, Room S-108, Fort Belvoir, VA 22060-5579 or usarmy.belvoir.hqda.mil.tioh-webmaster@mail.mil.)

2.3 Non-government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issue of the documents are those cited in the solicitation or contract (see 6.2).

ASTM INTERNATIONAL

ASTM – B487 - Metal and Oxide Coating Thickness by Microscopical Examination of a Cross Section , Measurement of
ASTM – E1282 – Chemical Composition, Specifying, and selecting Practice and Quantitative Analysis Methods for metal, and related Materials, Standard Guide for

(Copies for these documents are available online at https://www.astm.org)

SAE INTERNATIONAL

SAE-AMS-STD-595 - Colors Used in Government Procurement

(Copies of these documents are available online at https://sae.org)

American Society for Quality (ASQ)

ANSI/ASQ Z1.4 - SAMPLING PROCEDURES AND TABLES FOR INSPECTION BY ATTRIBUTES

(ASQ publications can be obtained from the American Society for Quality, 600 North Plankinton Avenue, Milwaukee, WI 53203-2914 or www.asq.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Tools. The government furnished hub or die must be used.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.3.

3.3 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs."

3.4 Materials. Materials shall conform to the referenced specifications and the requirements specified herein.

3.4.1 Copper base alloys. Copper base alloys shall be rolled, polished, free from pits, scale (including red oxide), dents, nicks, cracks, scratches, segregations and foreign inclusions that will not be removed in later processing. When tested as specified in 4.4.1, the chemical composition of the applicable copper base alloy shall be as specified in table II.

TABLE II. Unified Numbering System and Copper content of copper base alloys

UNS ALLOY NUMBER	ALLOY NAME	COPPER
C230	Red Brass	84.0% - 86.0%
C240	Low Brass	78.5% - 81.5%
C268	Yellow Brass	64.0% - 68.5%
C280	Free Cutting Brass	59.0% - 63.0%
C752	Nickel Silver (65-18)	63.0% - 66.5% **1/**

1/ When nickel silver is used as a base material, the nickel content shall be not less than 18% (C752).

3.4.2 Gold.

3.4.2.1 Gold alloy. When gold alloy is specified as a material of construction, the alloy shall be of the karat quality specified on the applicable specification sheet.

3.4.2.2 Gold for plating. Gold for plating shall be 24 karat.

3.4.3 Silver.

3.4.3.1 Sterling silver. Sterling silver shall be not less than 92.5% pure silver.

3.4.3.2 Fine silver. Silver shall be no less than 99% pure silver.

3.4.3.3 Silver alloy. Silver alloy when specified as a material of construction shall be of the fineness specified on the applicable specification sheet.

3.4.3.4 Silver for plating. Silver for plating shall be not less than 99.0% pure silver.

3.4.4 Nickel for plating. Nickel for plating shall be the nickel plating normally used in commercial

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practice.

3.4.5 Solder.

3.4.5.1 Soft solder. Soft solder shall be a lead-tin alloy having a melting point of not less than 375 °F (190.55 °C).

3.4.5.2 Hard solder. Hard solder shall be a silver-colored hard solder having a melting point of not less than 1075 °F (579.44 °C).

3.4.6 Epoxy. Epoxy for holding shanks in place shall be of a composition to sustain adhesion through 50 industrial dry-cleaning cycles.

3.4.7 Baking Enamel. The enamel for Type II, Style 3, Class B buttons shall be a flat, synthetic, baking enamel pigmented to the shade of black matching Color Chip Number 37038 of SAE-AMS-STD-595.

3.4.8 Lacquer. Lacquer shall be a pale clear synthetic lacquer. The use of a pigmented lacquer shall not be permitted except when a macublack oxidized finish is used.

3.5 Design. The embossed design on the buttons shall be an exact duplicate of the design of the Government loaned tools (See 6.3), from which the contractor's working die shall be extracted. The contractor's working die shall be tooled and polished to remove any dents, nicks, scratches, or other imperfections.

3.5.1 Soldering and swedging. All soldering shall be accomplished using hard solder specified in 3.4.5 except buttons with short, long and regular shanks may be soldered using soft solder specified in 3.3.4.2. Joints shall be clean, smooth, strong and free from burned or reduced areas. There shall be no excess solder, and all excess flux shall be removed. When hard solder is specified, hard solder specified in 3.3.4.2 shall be used. When tested as specified in 4.4.2.1, the shank shall show no break in the soldered connection or shall not become loose or wobbly. When tested as specified in 4.4.2.2, the soldered or swedged screw post shall not break loose or become wobbly. As an alternate, the Type I, Style 3 button shanks may be swedged by a cold headed process in lieu of soldering. The cold headed process shall require prior approval to ensure shanks are not loose or wobbly.

3.5.2 Stamping, trimming and piercing.- The embossed die shall be struck in a manner to ensure a well defined die struck outline. The button shall be trimmed and when applicable pierced to the die struck outline that are specified. All edges shall be clean smooth and free from burrs, drag, step and rough edges. The stamping and trimming and when applicable the piercing operation shall not damage or distort the design or alter the shape of the button.

3.6 Detail of components.

3.6.1 Type I (Uniform buttons).

3.6.1.1 Shells and liners. Except for oxidized silver buttons, all shells and when applicable liners, shall be fabricated from the red brass specified in 3.3.1. Shells for oxidized silver buttons shall be fabricated from nickel silver specified in 3.3.1. The thickness of the shells shall be as specified on drawing 4-4-92. When liners are required, the liners shall be 0.011 plus or minus 0.002 inch thick with 80 lines plus or minus 5 lines to the inch. Each liner shall have a raised platform to support the areas of embossed design on the shell and to ensure the proper positioning of the liner. The raised platform shall not be visible after assembly. Except for the raised embossed areas, style 2, class D shells shall

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be flat across the face. The rime of the face of the shell shall be burnished to produce a 30 to 40 degree angled bevel as shown on drawing 4-4-92.

3.6.1.2 Back and shanks. Backs and shanks shall be fabricated from low brass or red brass specified in 3.4.1. The thickness of the backs and shanks shall be as specified on drawing 4-4-92. Except for type I, style 2, classes A (20 ligne), D and E, type I, style 5, classes A, B and C, sized 22-1/2 and 35 ligne and type I, style 5, classes A and B, sizes 22-1/2, 35, and 40 ligne buttons, the type of shank shall be as specified (see 6.2). The type I, style 2, classes D and E buttons, the type I, style 4, classes A, B and C, sized 22-1/2, 35 and 40 ligne and type I, style 5, classes A and B, sizes 22-1/2 and 35 ligne buttons shall have a hopper back. All Type I, Style 3 buttons shall have a depressed back.

3.6.1.3 Assembly.

3.6.1.3.1 Pierced shells and liners. Pierced shells and liners shall be assembled with the lines of the liner parallel, within 5 degrees, for buttons larger than 25 lignes and 10 degrees for buttons 25 lignes or smaller, to the horizontal center line of the design on the shell and with the liner fitted against the back of the design on the shell within 1/32 inch. Further assembly shall be as specified in 3.6.1.3.2.

3.6.1.3.2 Shells, liners, button backs and shanks. Final assembly shall be as shown on drawing 4-4-92. The position of the shank, self-shank or hopper back shall be such that if a rod is inserted in the eye of either shank or self-shank or hopper back buttons, the rod shall be parallel to the lines of the back ground of the design of the 22, 22-1/2, 25 and style 3, 27 ligne buttons and at right angles to the parallel lines on the design of all other uniform buttons. A variation of 10 degrees maximum will be permitted.

3.6.1.3.3 Hopper back. When tested as specified in 4.4.8, the button shall not abrade or break the sewing thread.

3.6.1.4 Fastening devices. When an attaching device is required (see 6.2), the attaching device shall be as specified herein.

3.6.1.4.1 Toggles, split pin or ring. When split pin or ring type toggles are specified (see 6.2), the toggles shall be furnished detached. Split pin or ring type toggles shall be made from hard-drawn steel wire, zinc coated to withstand salt spray test specified in 4.4.3, for a period of 96 hours without indication of red rust. Spit pin or ring type toggles shall conform to the applicable design and dimensions shown on drawing 4-4-92.

3.6.1.4.2 Toggles, bodkin. When bodkin toggles are specified (see 6.2), the toggles shall be made from hard drawn nickel wire specified in 3.4.4, and shall have a nickel plated brass cap. The toggles shall conform to the applicable design and dimensions shown on drawing 4-4-92. The cap may be made from any type of brass.

3.6.1.4.3 Washers, button. Button washers, when specified (see 6.2), shall be nickel plated, yellow brass conforming to the applicable design and dimensions shown on drawing 4-4-92.

3.6.1.4.4 Links. Links for connecting matched pairs of uniform buttons, when required (see 6.2), shall be either chain or pin type as specified. The pin type link shall be furnished detached.

3.6.1.4.5 Link, chain. The chain link shall consist of six links conforming to the applicable design and dimensions shown on drawing 4-4-92, and formed from red brass specified in 3.4.1. The links

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shall not be soldered but shall be given a hot nickel plating after which a 24 karat gold electroplated finish (english finish) shall be applied. The breaking strength of the chain shall be not less than 15 pounds and shall not deform or break when tested as specified in 4.4.4.

3.6.1.4.6 Link, pin. The pin link shall be made from any type brass round wire with a 24 karat gold plated finish and shall conform to the applicable design and dimensions shown on drawing 4-4-92.

3.6.2 Type II (Cap buttons). Unless otherwise specified (see 3.6.2.4 and 6.2), all cap buttons shall be furnished with a screw tube.

3.6.2.1 Shells and liners. The shells and when applicable the liners shall be a specified in 3.6.1.1.

3.6.2.2 Backs, screw post and screw tube. Backs shall be made from low brass or red brass specified in 3.4.1, and the thickness of the back shall be as specified on drawing 4-4-92. The screw post and screw tube shall be made from free cutting brass specified in 3.3.1 and shall conform to the details and dimensions shown on drawing 4-4-92. All type II buttons shall have a regular back.

3.6.2.3 Assembly. Type II button shall be assembled as specified in 3.6.1.3, except the shank positioning requirements shall not apply. Screw posts shall be centered on and soldered or swedged to the back of the button.

3.6.2.4 Cap button for United States Coast Guard female officers. All Type II, Style 5, Class A, Subclass 2 USCG 22-1/2 ligne cap buttons shall have a staple type prong attaching device. Each prong of the attaching device shall be 7/8 inch ± 1/8 inch in length and shall be formed from 0.040 inch ± 0.002 inch diameter annealed gold-colored brass wire. The attaching device shall be centered on and fused, soldered, or swedged to the back of the button so that, when the prongs are spread open, they will be horizontal to the button design.

3.6.3 Type III buttons. Cuff link buttons shall be made from nickel silver specified in 3.4.3 and shall conform to the details and dimensions on drawing 4-4-92.

3.7 Finish.

3.7.1 Plating. All plating shall be by electroplating methods. The plating shall be nonporous and continuous over the entire plated surface. There shall be no cut-through, shaded, peeled, or blistered plating. Plating shall be smooth, fine grained, adherent, and free from pits, nodules, porosity, indications of burning, excessive edge build up, and other detrimental defects. When 24 karat gold and silver plating are tested as specified in 4.4.5, the average quantity of gold or silver on selected buttons shall be not less than the amount specified in TABLE III. In addition, no one dozen buttons tested shall contain less than 90% of the quantity of gold or silver specified in TABLE III.

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TABLE III Weight of gold or silver (grains) per dozen buttons

Ligne Size	Air Force Silver	Marine Corps Gold	Navy & Coast Guard Gold	Navy Silver	Coast Guard Auxiliary Gold	Coast Guard Auxiliary Silver	Army Solid Design Gold	Army Pierced Design Gold	Army Corps of Engineers Gold	Army JROTC Gold
20	6.0	--	--	--	--	--	1.0	--	--	--
20 1/	20.0	--	--	--	--	--	--	--	--	--
20 2/	--	--	--	--	--	--	--	--	1.5	--
20 3/	--	--	--	--	--	--	--	--	0.5	--
22	--	0.545	--	--	--	--	--	--	--	--
22-1/2 4/	--	--	1.0	7.5	1.0	4.5	--	--	--	--
25	12.0	--	--	--	--	--	1.0	--	--	--
25 2/	--	--	--	--	--	--	--	1.5	1.5	--
25 3/	--	--	--	--	--	--	--	0.5	0.5	--
25 5/	--	--	--	--	--	--	--	--	--	0.5
27	--	0.545	--	--	--	--	--	--	--	--
28	--	--	1.25	--	1.5	6.0	--	--	--	--
30	18.0	--	--	--	--	--	1.5	--	--	0.75
30 2/	--	--	--	--	--	--	--	2.25	2.25	--
30 3/	--	--	--	--	--	--	--	0.75	0.75	--
35	--	--	2.0	18.5	2.0	9.0	--	--	--	--
36	24.0	--	--	--	--	--	2.0	--	--	1.0
36 2/	--	--	--	--	--	--	--	3.0	3.0	--
36 3/	--	--	--	--	--	--	--	1.0	1.0	--
40	--	1.51	2.5	22.5	2.5	--	--	--	--	--
45	36.0	--	--	--	--	--	--	--	--	--

1/ Cuff link button
2/ Pierced design shell
3/ Pierced design liner
4/ When the button is to be procured for use on the service cap frame or should mark, the gold content specified shall not apply, however the button front, and rim shall be 24 karat gold electroplated.
5/ when the button is to be procured for use on the surface cap, the gold content specified shall not apply; However, the button front, and rim shall be 24 karat gold electroplated.

3.7.1.1 24 karat gold plating. The gold plating shall be accomplished using gold specified in 3.3.2.1. Prior to plating, all surfaces shall be cleaned of all matter that may affect the gold plating. The use of nickel as an undercoating with 24 karat gold shall not be permitted.

3.7.1.2 22 karat gold plating. The gold plating shall be accomplished using gold specified in 3.3.2.2. Prior to plating, all surfaces shall be cleaned of all matter that may affect the gold plating. The use of nickel as an undercoating shall be required.

3.7.1.3 Silver plating. Silver plating shall be accomplished using silver specified in

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3.3.3. The silver plating shall be a minimum of 0.0003 inch thick and shall be smooth, fine grained, adherent, free from visible blisters, pits, porosity, indications of excessive build up or other defects. In case of a dispute, testing shall be conducted in accordance with ASTM-B-487.

3.7.1.4 Nickel plating of buttons. Buttons shall be plated using not less than 0.0001 inch thick nickel plate specified in 3.3.7. Testing shall be standard commercial. In case of a dispute, testing shall be in accordance with ASTM-B487.

3.7.1.5. Nickel plating of washers, link, and toggles. Washers, links, and toggles (split pin, ring, or bodkin) shall be plated using not less than 0.0002 inch thick nickel plate specified in 3.3.7. Testing shall be standard commercial. In case of a dispute, testing shall be in accordance with ASTM-B-487.

3.7.1.6 Nickel underplating. When 22 karat gold plating or silver plating is specified, nickel underplating shall be used. The nickel underplating shall be not less than 0.00025 inch thick. Testing shall be standard commercial. In case of a dispute, testing shall be in accordance with ASTM-B-487."

3.7.2 Lacquering. Except for enameled and nickel plated buttons, the fronts of all buttons shall be lacquered using lacquer specified in 3.3.6. The dry lacquer film shall be hard, continuous, level, adherent, and free from lint, dust, and other foreign matter. When tested as specified in 4.4.6, the tissue paper shall not adhere to the clear lacquered surface

3.7.3 Enameling. When required, enameling shall be accomplished using enamel specified in 3.3.5. The baked enamel shall be hard, level, continuous adherent, uniform in color and free from dust or other foreign inclusions. When tested as specified in 4.4.7, the enamel shall not flake or peel.

3.7.4 Individual finish and color requirements. The colors and finishes of the buttons are as shown in TABLE IV. When deemed necessary, a standard sample, as described in 3.1, shall be furnished by the contracting office.

TABLE IV Finish.

BUTTON	FINISH
Type I:	
Style 1	
Class A	Silver plated, oxidized, relieved, satin finished, with a mirror-like burnished rim
Class B	Silver plated, oxidized, relieved, with a brushed satin finish on star, and a mirror-like burnished rim
Style 2	
Classes A B & C	Nickel plated, 22K gold plated, mirror-like polished, with a mirror-like burnished rim, and shall match TIOH Standard Metal Finish Chip Number 9 **1/**
Class D	24K gold plated, matte finished, polished highlights, with a mirror-like burnished rim
Class E	Bright nickel plated with a mirror-like burnished rim
Style 3, class A	24K gold plated, polished, with a mirror-like burnished rim
Style 4	
Class A	24K gold plated, matte finished, polished highlights, with a mirror-like burnished rim
Class B	Bronze, oxidized, relieved, satin finished
Class C	Silver plated, oxidized, relieved, satin finished, with a mirror-like burnished rim

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Class D	Nickel plated, 22K gold plated, mirror-like polished, with a mirror-like burnished rim, and shall match TIOH Standard Metal Finish Chip Number 9
Style 5	
class A	24K gold plated, matte finished, polished highlights, with a mirror-like burnished rim
class B	Bronze, oxidized, relieved, satin finished
Type II	
Style 1	Silver plated, oxidized, relieved, satin finished, with a mirror-like burnished rim
Style 2	
Classes A, B, & C	Nickel plated, 22K gold plated, mirror-like polished, with a mirror-like burnished rim, and shall match TIOH Standard Metal Finish Chip Number 9 **1/**
Class D	24K gold plated, matte finished, polished highlights, with a mirror-like burnished rim
Class E	Bright nickel plated with a mirror-like burnished rim
Style 3	
Class A	24K gold plated, matte finished, polished highlights, with a mirror-like burnished rim
Class B	Dull, flat, black enamel finished, and shall match Color Chip Number 37038 of FED-STD-595
Style 4	
Class A	24K gold plated, matte finished, polished highlights, with a mirror-like burnished rim
Class B	Bronze, oxidized, relieved, satin finished
Class C	Silver plated, oxidized, relieved, satin finished, with a mirror-like burnished rim
Class D	Nickel plated, 22K gold plated, mirror-like polished, with a mirror-like burnished rim, and shall match TIOH Standard Metal Finish Chip Number 9
Style 5	
Class A	24K gold plated, matte finished, polished highlights, with a mirror-like burnished rim
Class B	Bronze, oxidized, relieved, satin finished
Style 6	24K gold plated, matte finished, polished highlights, with a mirror-like burnished rim
Type III	24K gold plated, matte finished, polished highlights, with a mirror-like burnished rim

1/ At the option of the contracting office and/or the manufacturer, these buttons may be finished as follows: 24K gold plated, matte finished, polished highlights, with a mirror-like burnished rim.

3.8 Marking. The contractor shall stamp his trade-mark or other identifying marks legibly on the back of each button. The height of the mark shall not exceed 1/16 inch.

3.9 Workmanship. Buttons shall be clean, well made and shall meet the acceptable quality levels established by this specification. The button shall be free from burrs, fins, cracks, sharp or ragged edges and any defect which will affect appearance or serviceability.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection. Inspection and testing of the first article (see 3.2) shall be made of a completely finished item or items for all provisions of this specification applicable to the end product examination and tests (see 4.3.3 and 4.3.4).

b. Conformance inspection. Inspection shall be in accordance with the provisions set forth herein, except where otherwise indicated (see 6.2). Sampling for inspection shall be performed in accordance with ANSI/ASQC Z1.4.

4.2 Inspection conditions. Unless otherwise stated, all inspections shall be performed in

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accordance with the test conditions specified in 4.3.

4.3 Methods of inspection.

4.3.1 Testing of components. See TABLE V, below:

TABLE V Components testing.

COMPONENT	CHARACTERISTIC	RQM'T¶	TEST METHOD	RESULTS REPORTED AS
copper based alloy	chemical composition	3.41	4.4.1	nearest 1%

4.3.1.2 Certification of compliance. Materials listed below may be accepted on the bases of a contractor's certification of compliance for requirements specified in the applicable paragraphs of the specification.

Component	Characteristic	Rqm't para
Gold for plating	Karat	3.3.2
Silver for plating	Fineness	3.3.3
Solder	Material identification	3.3.4
Enamel (organic)	Material identification	3.3.5
Lacquer	Material identification	3.3.6
Shell back and liners	Thickness	3.6.1
Split pin or ring type toggles	Material identification and type of coating	3.6.1.4.1
Bodkin toggle	Temper of wire and material identification of cap	3.6.1.4.2
Pin link	Material and finish identification	3.6.1.4.3

4.3.2 In-process inspection. Inspection shall be made at any point or during any phase of the manufacturing process to determine whether operation or assemblies are as specified. The Government reserves the right to exclude from consideration for acceptance any material or service for which in-process inspection has indicated nonconformance. In-process inspection shall be conducted to see that accomplishment of the following is in accordance with the specification requirements.

Requirement, operation or assembly	Characteristics	Rqm't para
Gold plating	White metal underplating	3.7.1.1

4.3.3 Inspection of the end item.

4.3.3.1 Visual examination of buttons. Examination shall be made at a distance of approximately 16 to 18 inches. The defects found during examination shall be classified in accordance with 4.3.3.1.1, 4.3.3.1.2 and 4.3.3.1.3. The lots shall be inspected in accordance with 4.3.3.1.1. The unit of product for the examinations in 4.3.3.1.1, 4.3.3.1.2 and 4.3.3.1.3, shall be one completely fabricated button and on toggle, split pin, ring, chain link, pin type link, washer and screw tube, as applicable.

4.3.3.1.1 Examination of buttons for defects in finish, design, material, construction, workmanship and markings. Defects designated by an asterisk (*) shall be classified as major when seriously affecting appearance or serviceability and minor when affecting appearance or serviceability but not seriously.

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TABLE VI Classification defects

Examine	Defects	Classification		
		Major	(*)	Minor
Color and finish	Not as specified.....	x		
	Does not compare favorably with standard or approved sample.....	x		
Button front	Any plated area broken, cut-Through, peeled or blistered.....	x	-	-
	Discoloration, stain or speck.....	-	*	-
Lacquer	Missing when required.....	x	-	-
	Hazy, cloudy or powdering.....	-	-	x
	Foreign matter imbedded.....	-	-	x
	Not smooth, not uniform, not continuous or not adherent, i.e. flaking, blistering, peeling or has runs, sags, or drops.....	-	*	-
Enamel	Missing.....	x	-	
	Not adherent.....	x	-	
	Burnt.....	-	*	
	Bubbles, pits or cracks.....	-	*	
	Inclusions of foreign matter.....	-	*	
	Stained or darkened.....	-	*	
Design	Detail altered, does not conform to Government hub or standard sample.....	x		
	Warp, twist or distortion producing irregular surface contour or outlined.....	x		
	Any significant detail obliterated.....	x		
	Any significant detail reduced, i.e, does not compare favorably with approved sample.....		*	
	Evidence that grooved lines per inch of liners is not within specified tolerance (measure when in doubt).....			x
	Not type, style, class, subclass or size specified.....	x		
Metal	Surface spotted or open grained, i.e., appearing as pitted, porous or crystalized.....	x		
Piercing	Not pierced where specified.....	x		
	Edges not clean cut and smooth; not free from burrs, drag, step and rough edges.....	-	*	
Assembly of pieced design button	Liner not positioned properly, i.e., parallel lines off horizontal more than 5 degrees, for buttons larger than 25 ligne and more than 10 degrees for buttons 25 ligne of less.....	x		
	Liner not fitted against back of design on shell within specified tolerance.....		*	
Workmanship	Metal mark, such as, dig, dent or scratch on polished surface or in background, noticeable at 16 inches.....	-	*	
	Pinhole, crack or rupture.....	x		

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	Solder spatter on front……	x		
Assembly	Sharp edge, burr or fin appearing on front, edge or		*	
	Classification defects continue			x
	back of button……			
	Rolled edge not flush with back by more than 1/32 inch	-	-	x
	at any point……	x	-	-
Shank	Looseness; shell edge not	x	-	-
	closed in, securely clinching the back. (Rotation of	x		
	back within closed shell shall not exceed 10		*	
	degrees maximum.)……		*	
	Edge cracked or split……..			
	Missing, out of position, or is not type specified……..			x
	Eye of hopper back not formed as required…….		*	
	Bent, loose, deformed or otherwise impaired……..			
Threaded post	Solder spatter on back exceeding 1/8 inch in maximum	-	*	
	dimension……..	x		
	Sharp edge, fin or burr in eye of hopper back……		*	
	Eye of shank improperly positioned by more than 10	x		
	degrees……	-		x
	Missing……..			
Link: pin,	Bent, loose, deformed or otherwise impaired……		-	x
chain and cuff	Thread stripped, flattened or damaged……	-	*	
	Not as specified……	x		
	Solder spatter on back exceeding 1/8 inch in maximum	x		
	dimension……..	-	-	x
Toggles,	Deformed, twisted, bent or otherwise damaged…….	-	*	
bodkin	Not specified type……	x	-	-
	Missing……..	-	-	x
	Not attached to button (except pin type link)……..	-	*	
	Sharp edge, fin or burr……	-	*	
	Toggle or cap missing…….	-	*	-
Marking	Not type specified…….	-	-	x
	Cap loose……..	-	-	x
	Deformed, bent, twisted or otherwise damaged…….	-	-	x
Staple type	Sharp edge, fin burr…….	-	-	x
prong	Not attached to button…….	-	-	x
	Missing or not as specified…….	-	-	x
	Illegible or misspelled……..	-	-	x
	Misplaced or incorrect size…….			
	Not material specified……..			
	Not properly swedged or soldered…….			
	Not annealed……			

4.3.3.1.2 Visual examination of unattached components (toggles, split pin or ring, chain link, pin type link, washer, button and screw tubes, as applicable. The components which are not attached to the buttons shall be examined in accordance with the classification of defects shown in table VI and table VII.

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TABLE VII Classification of unattached components.

Examine	Defects
Screw tube	Thread stripped, flattened or damaged. Not type specified. Opening obstructed, will not pass threaded post of corresponding size. Bent, damaged or deformed affecting serviceability. Bent, damaged or defective affecting serviceability. Deformed, bent twisted or otherwise damaged.
Washer, button	Not type specified.
Toggles, split pin or ring, type link	Sharp edge, burr or sliver. Loose fit to shank, i.e., may become detached in use. Attached to button.
Examine	
Markings, exterior and interior	Missing, incorrect, illegible or improper size, location, sequence or method of application.
Materials	Component missing, damaged or otherwise defective, affecting serviceability.
Workmanship	Inadequate application of components such as incomplete closure of envelope and container flaps, insecure seating of fastening device, tissue wrapping incomplete, voids inadequately filled and building or distortion of containers.
Weight	Net weight exceeds the requirements.

4.3.3.1.3 Examinations of buttons and components for defects in dimensions. Any dimension that is not within the specified tolerance shall be classified a defect.

4.3.4 End item testing. Testing of the completely fabricated button shall be performed in accordance with table VIII for the characteristics shown therein.

TABLE VIII Testing of the end item.

Characteristic	Rqm't para	Test Method	Rqm't appl to Individual	Rqm't appl to Lot Unit average	Sample unit	Results reported as
Assembly: Hopper back	3.6.1.3.3	4.4.8	x		1 button	Pass or fail
Toggles, split pin or ring resistance to salt spray	3.6.1.4.1	4.4.3	x	-	3 each	Pass or fail
Link chain, tensile strength	3.6.1.4.5	4.4.4	x		1	Pass or fail

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Plating: Silver plating	3.7.1.1 & table III	Std comm'l 4.4.5.2 (in case of dispute)	x -	- X	12 buttons	Nearest 0.1 grain
Gold Plating:	3.7.1.1 & table III					
Shell		Std comm'l 4.4.5.1 (in case of dispute)	x -	- x	12 buttons	Nearest 0.1 grain
Liner		Std comm'l 4.4.5.1 (in case of dispute)	x	x	12 buttons	Nearest 0.1 grain
Nickel plating: Identification	3.7.1.2.1	Std comm'l	x		1/	Pass or fail
Thickness	3.7.1.2.2	Std comm'l	x	-	1/	Nearest 0.00001 inch
Lacquer tests: Liver of sulfur	3.7.2	4.4.6.1	x	-	1 button	Pass of fail
Soldering (shank)	3.5.1	4.4.2.1	x	-	1 button	Pass of fail
Soldering or swedging (screw post)	3.5.1	4.4.2.2	x	-	1 button	Pass of fail
Enamel adherence test	3.7.3	4.4.7	x	-	1 button	Pass of fail

1/ Sample unit shall be one button, attachment or component.

4.4 Tests.-

4.4.1 Chemical composition test. Chemical composition shall be determined in accordance with test method III of FED-STD-151. Results shall be evaluated to determine compliance with the requirements in 3.3.1.

4.4.2 Tests for soldered and swedged connections.

4.4.2.1 Tests for shank type buttons. The shank type button, except self shank and hopper back buttons, shall be placed in an appropriate testing device so that the shank is maintained in a vertical position. A steadily increasing compression load shall be applied directly up the upright shank until the eye is noticeably elongated. Remove the compression load and examine the joint for failure to determine conformance with 3.5.1.

4.4.2.2 Test for screw post type buttons. The screw post type button (soldered or swedge button back), shall be placed in an appropriate testing device so that the screw post is maintained in a

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vertical position. A steadily increasing tension load to 50 pounds shall be applied directly upon the upright post. Remove the button from testing device and examine joint to determine conformance with 3.5.1

4.4.3 Salt spray test. The salt spray test shall be in accordance with test method 811 of FED-STD-151, to determine conformance with 3.6.1.4.1.

4.4.4 Chain link test. The chain shall be placed in a suitable testing device and a d steadily increasing direct tension load up to 15 pounds shall be applied to the chain to determine conformance with 3.6.1.4.5.

4.4.5 Test for gold and silver content. Standard commercial laboratory test methods shall be used to determine gold and silver content conformance with 3.7.1. In case of gold and silver content dispute, the procedure specified in 4.4.5.1 and 4.4.5.2, as applicable shall be used.

4.4.5.1 24 karat gold content. Boil 12 shells or 12 liners, as applicable, with 60 ml. 1:1 nitric acid (HN03) in a 400 ml. beaker using a small flame or hot plate. Continue boiling until all metals, except gold, are in solution. Dilute to 200 ml. with hot water. Filter through a rapid ashless filter paper and wash several times with hot water. Take the filter paper with fold (cutting off any excess filter paper), wrap in a 2x2 inch piece of silver-free lead foil, and roll into a ball. Place cupel in a muffle furnace for one hour at 1,000 degrees C. Remove from the furnace, and examine hollow for flakes of gold, impurities and discolored gold bead. If any of these conditions are noted, add additional lead shot and return cupel to the muffle furnace for one hour. Otherwise, cool the gold bead and weigh.

4.4.5.2 Silver content. Boil 12 shells or 12 Air Force cuff link buttons, as applicable, with 60 ml. 1:1 nitric acid (HN03) in a 400 ml. beaker using a small flame or hot plate. Continue boiling until all metals are in solution. Dilute in volumetric flask to 1,000 ml. Pipette 200 ml. aliquot from the flask into a 400 ml. beaker. Heat to boiling and add HCL (3:100) drop-by-drop from a pipette until precipitation is complete (1 ml. of HCL solution precipitates about 0.038 grams of silver). When solution has cooled and precipitate has settled, determine whether precipitation is complete by adding a few drops of the HCL (3:100). If solution remains clear, filter on a weighted porcelain Gooch crucible. Wash the precipitate server time with HN03 (1:200) and finally wash with cold water. Protect the silver chloride precipitate from the light. Dry in an oven for not less than two hours, cool and weight the silver chloride. The silver content shall be calculated as follow:

Quantity of silver/doz.
Buttons in grains + weight AgCL x 15.4324 x .7526 x 5

4.4.6 Test for lacquer. At room temperature (60° to 80° Fahrenheit/15.6° to 26.7° Celsius), press a piece of tissue paper against the lacquered surface for 15 seconds, using any pressure capable of being exerted between thumb and two fingers, after which the pressure shall be released and the item inspected to determine compliance with 3.7.2.

4.4.7 Adherence test. A sharp, pointed instrument shall be drawn or wiggled around the beveled edge of the button or across a similar flat polished section, cutting through the enamel to the metal. During this test observe the enamel coating to determine conformance with 3.7.3.

4.4.8 Test for hopper backs. Hopper back buttons shall be hand sewn to any cloth material with four double cotton threads conforming to Type I or Type II, 4 ply of A-A-52094. This button shall be twisted 180 degrees in opposite directions, five times in each direction and then examined to determine compliance with 3.6.1.3.3.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6.NOTES

6.1 Intended use. The buttons covered by this specification are intended to be worn on certain uniforms of the United States Military Services.

6.2 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type, style, class, subclass (when applicable) and size of button required (see 1.2).
(c) When standard samples will not be furnished (see 3.1).
(d) When first articles are required (see 3.2).
(e) When a government hub will not be furnished (see 3.1)
(f) When the contractor is required to furnish his dies to the Government upon completion of the contract (see 3.1).
(g) The type of back and when applicable the length of shank required (see3.6.1.2).
(h) The type of attaching device required, when applicable (see3.6.1.4).
(i) When a screw tube will not be required with cap buttons (see 3.6.2 and 3.6.2.4).

6.2.1 Contract data requirements. Data conforming to Department of Defense data item description DI-R-4803, DI-R-4805, DI-T-4901, DI-T-4902, DI-T-4903 and DI-T-4904 will usually be required for delivery in connection with this specification for Marine Corps procurements. For other than Marine Corps procurements, the cognizant procuring activity will provide comparable data item descriptions. When so required, such data will be specified for delivery on a DD-1423 included in the contract.

6.3 Coast Guard. The Coast Guard and Coast Guard auxiliary standard samples (see 3.1) may be obtained by addressing the following: “C commanding” Officer, U.S. Coast Guard Supply Center (Attn: Code 310), 830 Third Avenue, Brooklyn, New York 11232.

6.4 Army gold plated (20 ligne) solid design button. Standard samples of the army gold plated 20 ligne solid design button may be obtained by addressing the following: Director, The Institute of Heraldry, U.S. Army, Attn: DAAG-HDT, Cameron Station, Alexandria, Virginia 22314.

6.5 Hopper back button. Army and Air Force hopper back buttons have been designed as sew-on type only and will replace the long, regular and short shank as the standard. These buttons are not to be used with toggles, bodkins or rings. The Navy hopper back button is designed to be both a sew-on type and a bodkin secure type, where applicable.

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6.6 Army standard buttons. The standard button for the Army will be a solid design button. Pierced design buttons with the various backs may be sold commercially but will not be Government procured. The Army Class E buttons are restricted to use by the U.S. Army Band and Ceremonial units of the 3rd Infantry.

6.7 CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

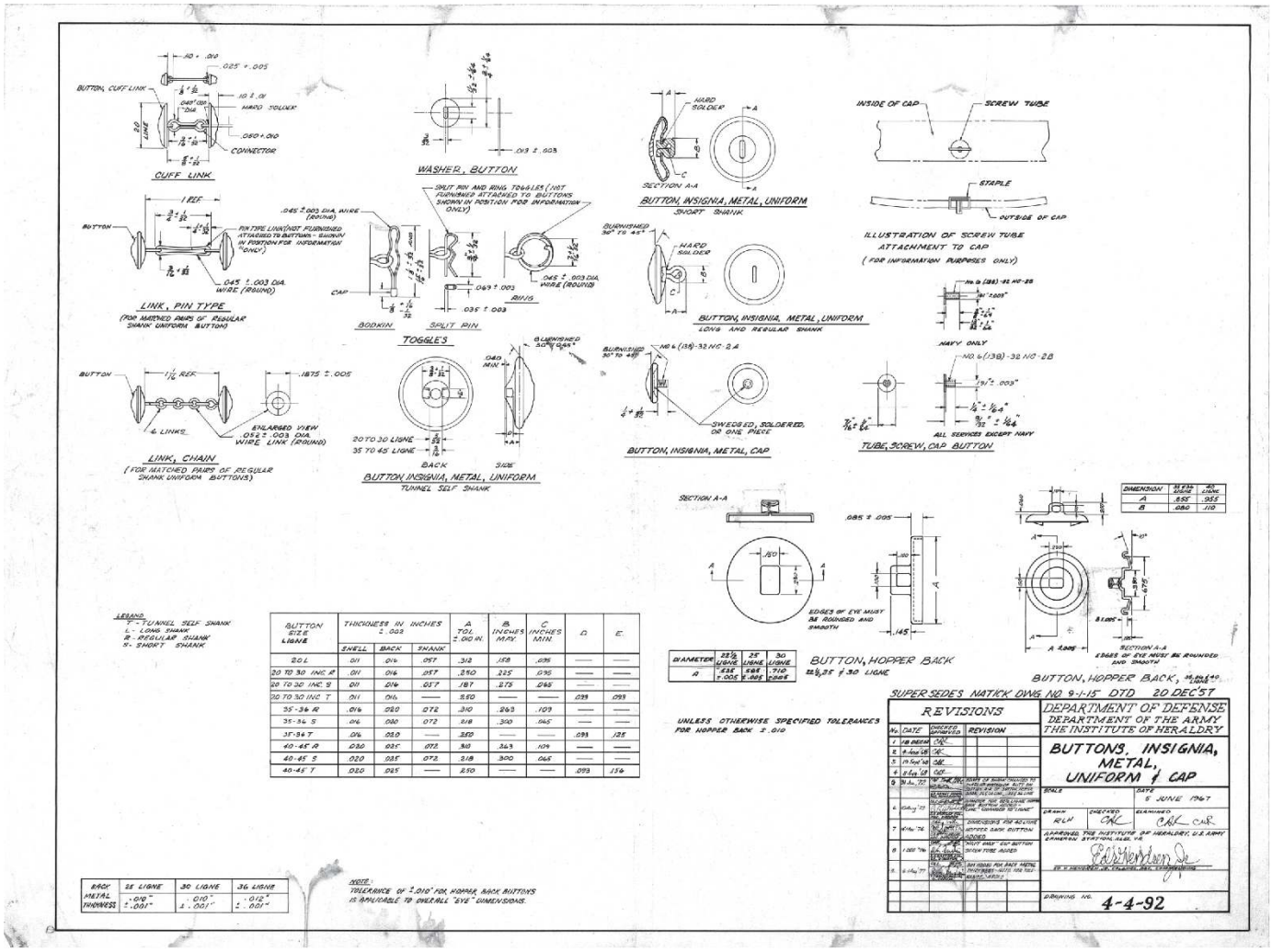


Figure 1 Buttons Insignia Metal, Uniform & Cap

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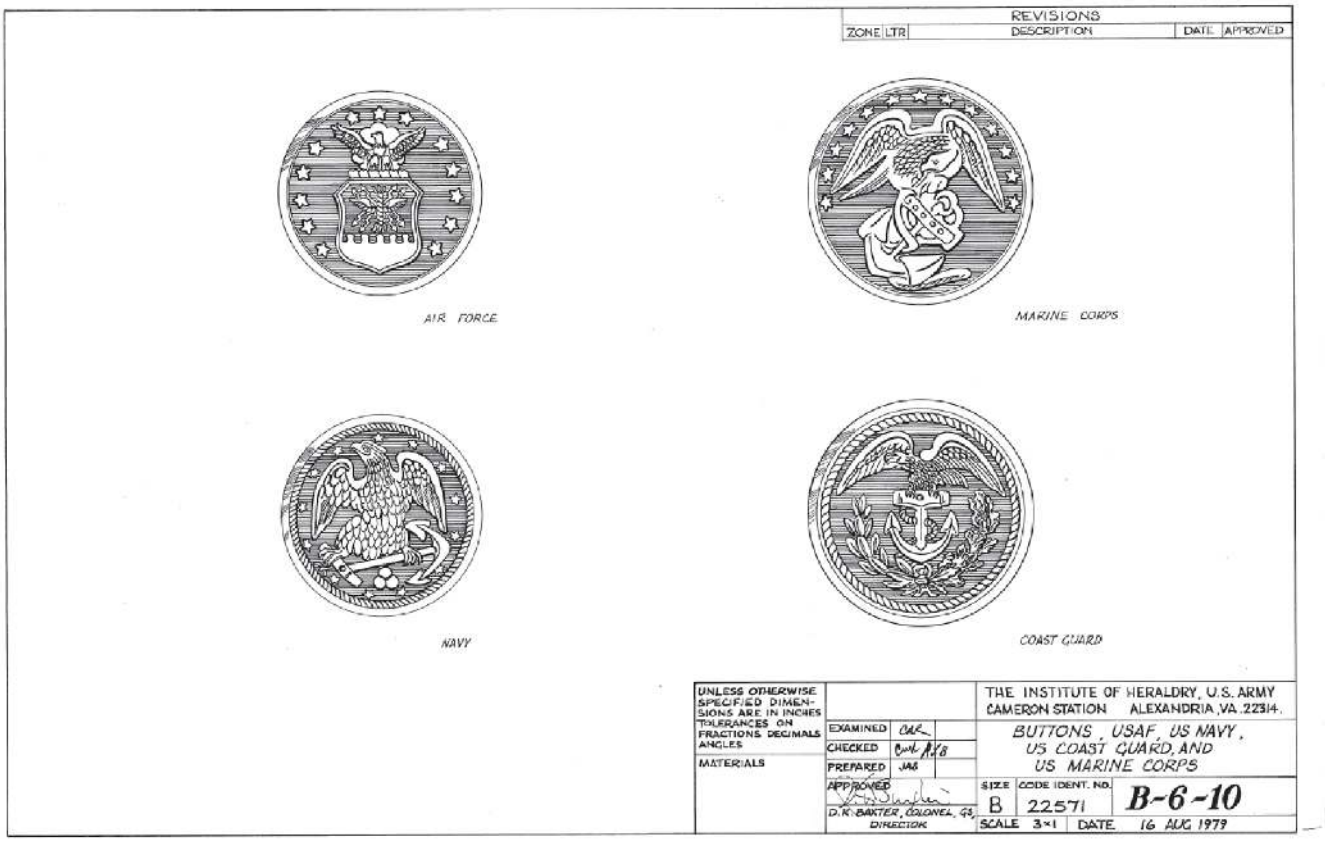


Figure 2 Buttons USAF, US Navy, US Coast Guard, and US Marine Corps

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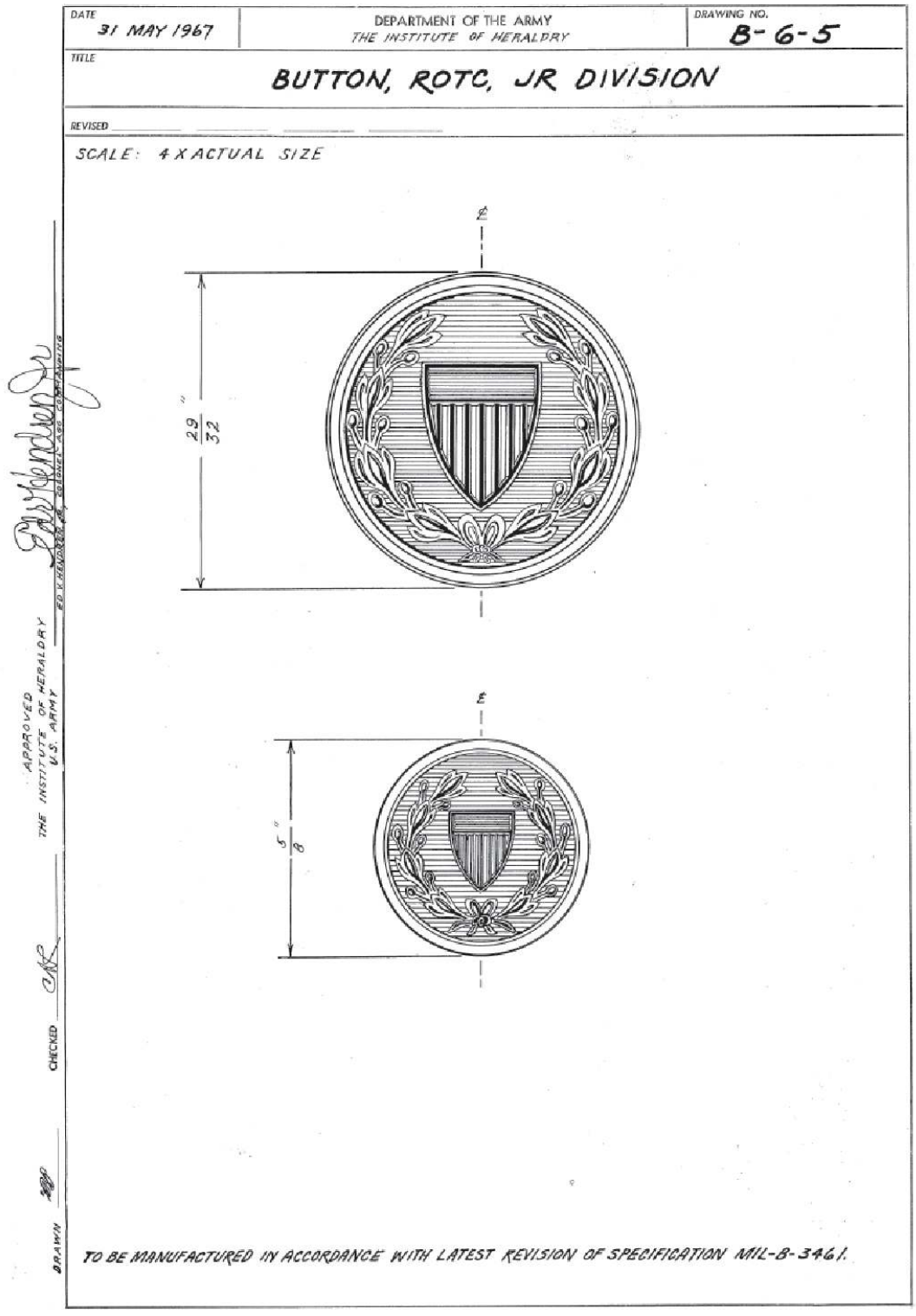


Figure 3 Button, ROTC, Jr Division

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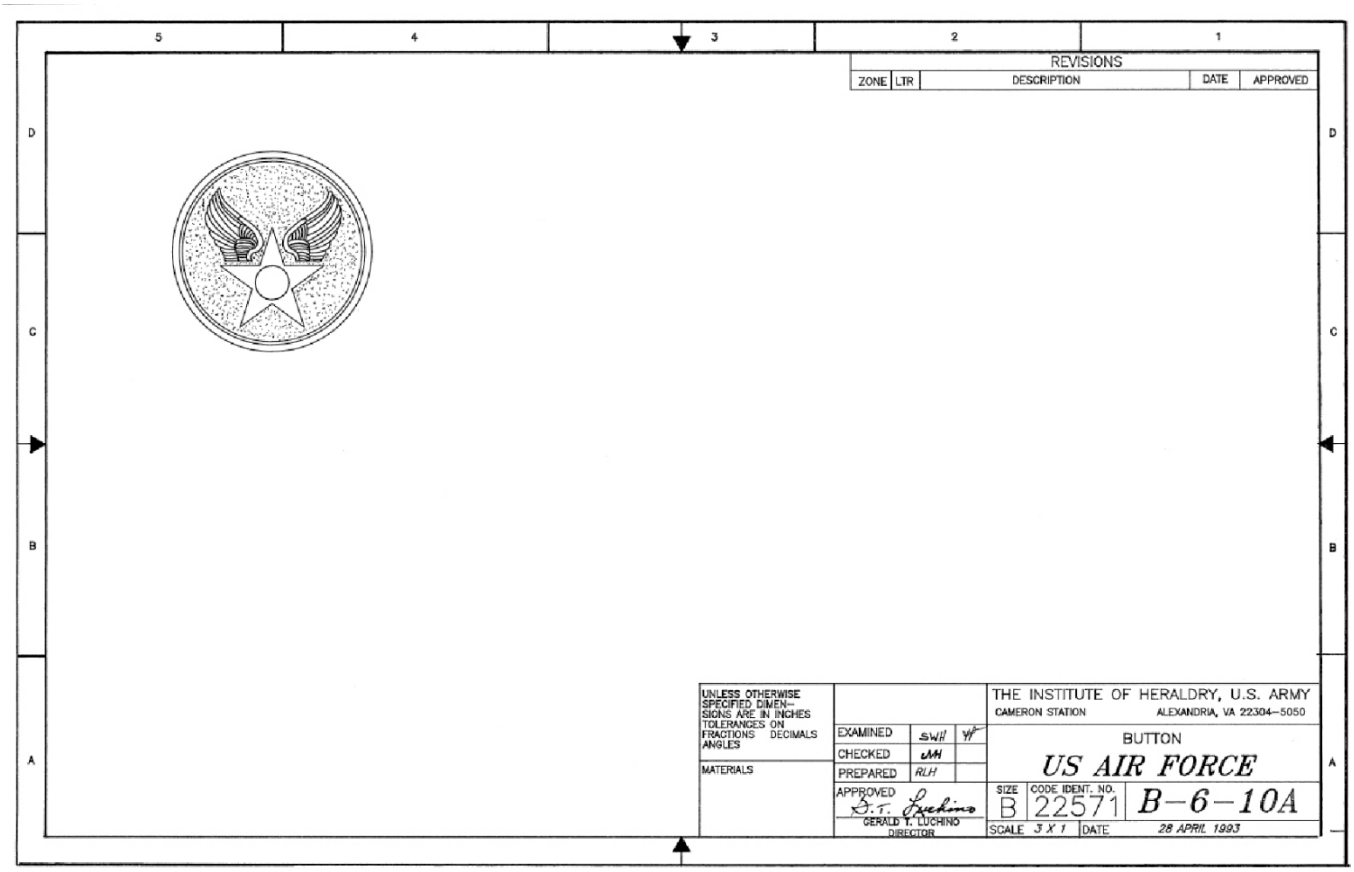


Figure 4 US Air Force

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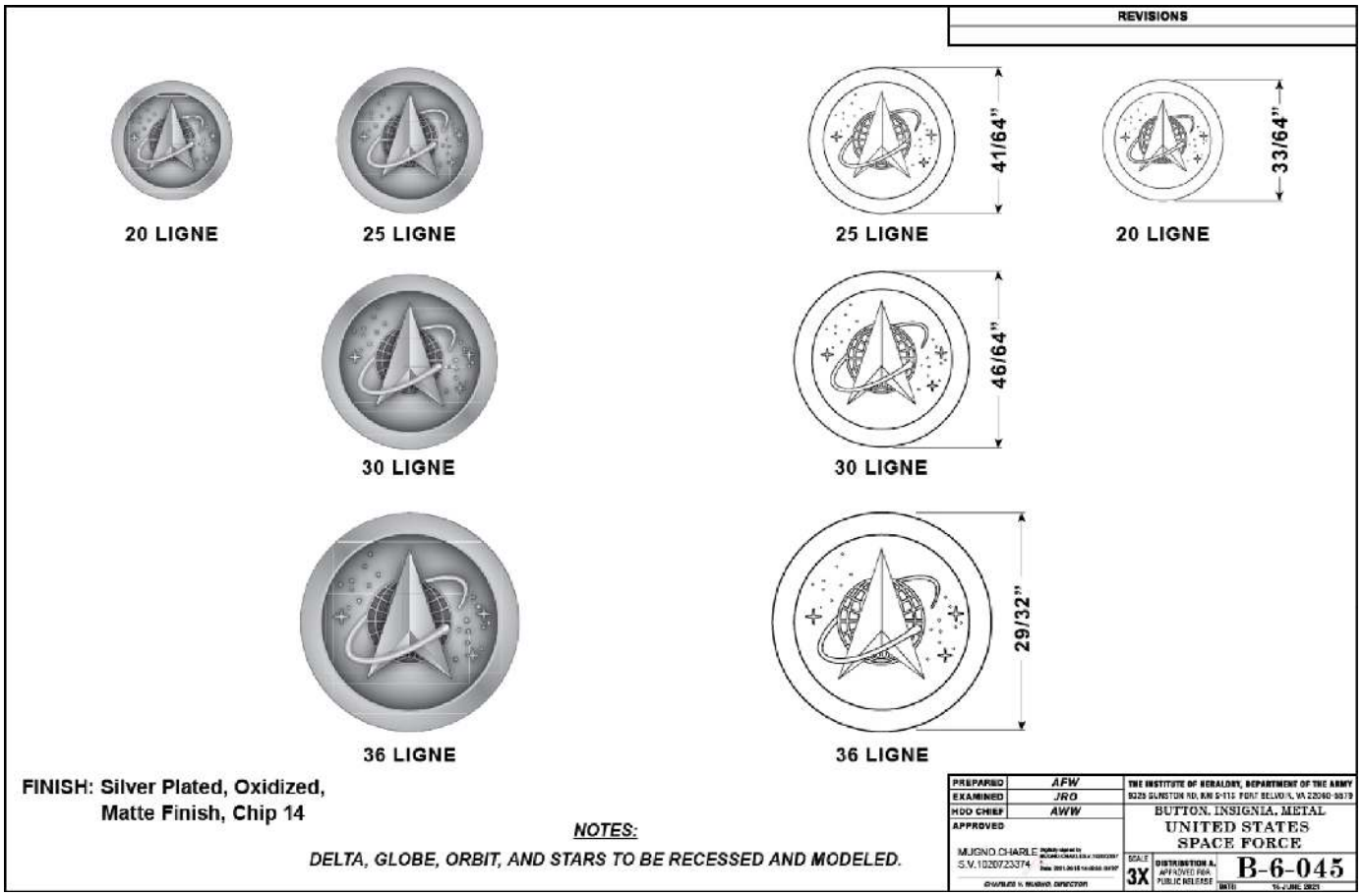


Figure 5 US Space Force

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Custodians:
Army - IH
Air Force - 11
Navy - NU

Preparing Activity:
Army - IH

Review Activities:
Navy - MC, CG1
DLA - CT

(Project No. 8455-2025-076)

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